ANICET NZEUKOC

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SUMMARY

I am a PhD candidate in Economics at the University of Montreal and a member of CIREQ, supervised by Professors Marine Carrasco and René Garcia. My research fields are econometrics and financial econometrics: functional data analysis, structural breaks, option pricing and risk-neutral densities, and machine learning methods for structural models. I develop instrumental-variable and change-point methods to recover and monitor the probability distributions that option markets assign to future asset prices. I bring 5+ years of experience in quantitative research across Cameroon and Canada, and I work in French and English.

EDUCATION

University of Montreal (UdeM), CIREQ *2022–*
PhD candidate in Economics, Canada. Supervisors: Prof. Marine Carrasco and Prof. René Garcia

University of Yaoundé II *2021–2022*
MSc in Economics, Cameroon

Sub-regional Institute of Statistics and Applied Economics (ISSEA) *2018–2021*
Engineer in Statistics and Economics (ISE), Cameroon

University of Dschang *2016–2018*
MSc in Applied Mathematics, Cameroon

University of Dschang *2013–2016*
BSc (Licence) in Mathematics and Computer Science, Cameroon

PROFESSIONAL AND RESEARCH EXPERIENCE

Télé-université (TELUQ) April 2026 – August 2026
Research Assistant (part-time) *Montréal*

- Governance, directors' and officers' insurance and executive compensation.
- Built a firm-year panel on more than 220 Canadian public firms (2013–2024) with an automated extraction pipeline in Python.
- Applied difference-in-differences and event-study methods.

HEC Montréal December 2022 – August 2025
Research Assistant, with Prof. Pierre-Olivier Pineau *Montréal*

- Stochastic frontier analysis of energy efficiency across the ten Canadian provinces.
- Drafted "Measuring Provincial Energy Efficiency in Canada: A Stochastic Frontier Analysis".

Scikoop Inc. / IVADO January 2023 – June 2023
Research Consultant, IVADO Scientist in Residence program *Montréal*

- Statistical tools for equity, diversity and inclusion in public-health and social-science research.

ISSEA October 2021 – July 2022
Research Assistant, with Dr. Ignace Kamga *Yaoundé*

- Macroeconomic forecasting for the CEMAC countries.
- Organised the weekly "Research Wednesday" seminar.

Teaching *2024–2025*
Lecturer, Econometrics I (UdeM, ECN1260); Fall 2024
Lecturer, Principles of Economics (UdeM, ECN1000); Fall 2025

Teaching assistant *2021–*
Economic Data Analysis (UdeM, ECN1160); Fall 2025 and Fall 2026, for Prof. Marine Carrasco
Advanced Econometrics Theory (UdeM, ECN7060, PhD); Winter 2026, for Prof. Marine Carrasco
Numerical Methods in Economics (UdeM, ECN6338); Winter and Fall 2025, for Prof. William McCausland
Techniques of Economic Analysis II (UdeM, ECN1075); Summer 2025, for Prof. Eugène Deta'a
Econometrics I (UdeM, ECN1260); Winter 2025, for Prof. Amal Serre
Applied Econometrics (UdeM, ECN2260); Winter 2024, for Prof. William Arbour
Microeconomic Theory I (UdeM, ECN2040); Winter and Fall 2023, for Prof. Andriana Bellou
Microeconomics (ISSEA, engineering-economist students); 2021–2022, for Dr. Ignace Kamga

Research in progress
"Instrumental Variable Estimation of Risk-Neutral Density using Functional Linear Models," with Marine Carrasco and René Garcia.
"Deep Surrogates in Option Pricing Models for Risk-Neutral Density Estimation," with Rostand Tchouakam.
"Detecting and Locating Structural Changes in a Sequence of Estimated Risk-Neutral Densities," work in progress.

AWARDS AND FELLOWSHIPS

Fonds de recherche du Québec – Société et culture (FRQSC), Doctoral Research Fellowship (B2), 2026–2027
First Prize, Poster Competition, 90th ACFAS Conference, Université de Montréal, 2023
Doctoral Admission Fellowship, Department of Economics, UdeM, 2022–
Tuition Fee Exemption Scholarship, University of Montreal, 2022–
Government of Cameroon Admission Scholarship, ISSEA, 2018–2021

AFFILIATIONS

Centre interuniversitaire de recherche en économie quantitative (CIREQ), 2023–
Society for Financial Econometrics (SoFiE), 2024–

TECHNICAL STRENGTHS

Programming	Python (NumPy, Pandas, PyTorch, QuantLib, scikit-learn), R, MATLAB, Stata, SQL, Git
Data	OptionMetrics Ivy DB, large-scale time series, parallel computing, reproducible pipelines
Office	LaTeX, Excel (advanced), PowerPoint

VOLUNTEERING ACTIVITIES AND TALKS

Chair, Quebec PhD Workshop in Economics, Statistics and Finance (QPESF) *2022–*
Organiser and moderator of a weekly doctoral seminar in economics, statistics and finance across

Quebec universities.

Co-organiser, CIREQ Doctoral Conference on Economics and Finance *April 2026*
Organising committee with Prof. Marine Carrasco (UdeM), Jackson Montelpare (McGill) and Zahraa Arkahdan (Concordia).

Volunteer, Cap Campus (Université de Montréal) *2024–*
Workshops in applied mathematics, finance and statistics for high-school and college students.

Talks *2023–2026*
41st Annual Meeting of the Canadian Econometrics Study Group (CESG), Simon Fraser University, Vancouver; October 2026 (poster)
65th Annual Congress of the Société canadienne de science économique (SCSE), Québec City; May 2026
CIREQ Doctoral Conference on Econometrics and Finance, Université de Montréal; April 2025
Doctoral seminar, Department of Economics, Université de Montréal; May 2025 and December 2025
Inclusive Society Colloquium, Université Laval, Québec; June 2023
ACFAS Conference, 90th edition, Université de Montréal; May 2023 (poster)

LANGUAGE AND OTHER INTERESTS

French (native), English (fluent), football, reading, writing

ABSTRACT OF SOME SELECTED WORK IN PROGRESS

Instrumental Variable Estimation of Risk-Neutral Density using Functional Linear Models
With Marine Carrasco and René Garcia
Nonparametric estimation of the risk-neutral density from a time series of European option prices. Pooling contracts across dates enlarges the sample but makes the strike endogenous, because the exchange sets strikes in response to the same shocks that move prices. The paper develops an instrumental-variable sieve estimator in Hermite basis functions, using lagged values of the index and of the VIX and SKEW indices as instruments, together with an estimator-free confidence band for the entire density that accommodates serially dependent pricing errors. The Monte Carlo design is the estimated model written in reduced form, so the experiment reproduces the measured degree of endogeneity. Applied to S&P 500 options, OptionMetrics, 1996–2022.

JEL Classification: C14, C26, C58, G13
Keywords: Risk-Neutral Density, Endogeneity, Instrumental Variables, Sieve Estimation

Deep Surrogates in Option Pricing Models for Risk-Neutral Density Estimation
With Rostand Tchouakam

Detecting and Locating Structural Changes in a Sequence of Estimated Risk-Neutral Densities
Sole-authored